



A hybrid seasonal autoregressive integrated moving average and quantile regression for daily food sales forecasting



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ABSTRACT

In the retail stage of a food supply chain, food waste and stock-outs occur mainly due to inaccurate forecasting of sales which leads to incorrect ordering of products. The time series sales in food retail industry are characterized by high volatility and skewness, which vary by time. So, the interval forecasts are required by the retail companies to set appropriate inventory policy (reorder point or safety stock level). This paper attempts to develop a seasonal autoregressive integrated moving average with external variables (SARIMAX) model to forecast daily sales of a perishable food. The process of fitting a SARIMAX model in this study involves: (i) the development of Seasonal Autoregressive Integrated Moving Average (SARIMA) model and (ii) combining the SARIMA model and the demand influencing factors using linear regression. As the SARIMAX using multiple linear regression (SARIMA-MLR) model produces only mean forecast, the possibility of underestimation and overestimation is very high due to high service level, peak, and sparse sales in food retail industry. Therefore, a hybrid SARIMA and Quantile Regression (SARIMA-QR) is developed to construct high and low quantile predictions. Instead of extrapolating the quantiles from the mean point forecasts of SARIMA-MLR model based on the assumption of normality, the SARIMA-QR model directly forecasts the quantiles. The developed SARIMA-MLR and SARIMA-QR models are applied in modeling and forecasting of sales data, i.e., the daily sales of banana from a discount retail store in Lower Bavaria, Germany. The results show that the SARIMA-MLR and -QR models yield better forecasts at out-sample data when compared to seasonal naïve forecasting, traditional SARIMA, and multi-layered perceptron neural network (MLPNN) models. Unlike the SARIMA-MLR model, the SARIMA-QR model provides better prediction intervals and a deep insight into the effects of demand influencing factors for different quantiles.

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1. Introduction

In food retail industry, accurate and timely sales forecasting plays a major role in profitable operation and better stock management. The retail food stock comprises huge varieties of perishable foods with short shelf-lives and different storage conditions, which makes the food sales forecasting a complex task (Mena et al., 2014). Therefore, if a retail manager forecasts the food sales correctly that can lead to better customers' satisfaction, increase in sales, and food waste reduction. Eriksson (2012) and Buzby et al. (2009) also suggested the improvement of forecast and ordering system to reduce in-store retail food waste. In general, the forecasting in food retail industry can be done qualitatively, quantitatively or combination of both. The qualitative methods are subjective or judgmental assessments, which are

applied when there is no sufficient data to do a quantitative assessment methods. The quantitative methods involve either the development of mathematical models with causal variables or extrapolation of historical sales data. The quantitative methods are further classified into time series models, causal models and hybrid models. The causal models rely on cause–effect relationships between sales and demand influencing factors. The time series models rely on historical sales time series data observed at regular intervals to predict the future demand. The hybrid models are combination of both time series and causal models.

The accuracy of forecasting in food retail industry highly depends on the forecast horizon (Sickle and Beilock, 1981). The shelf-life of product is an important criteria in identifying the time horizon to forecast. Especially, fresh fruits and vegetables which have short shelf-life, require short-term forecasting in order to avoid redundant or insufficient stocks (Armstrong, 2001a). In this case, the daily forecasting is highly advantageous, since aggregation of daily forecasts (for one or more days) can provide better

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results than the forecasts from aggregated data (Dekker et al., 2004). The demand patterns in food retail are product and customer specific. The variation in demand can be categorized into short-term fluctuations (holidays, promotions), medium seasonal patterns (school vacation, season) and long-term trend (economic situations) (Žliobaitė et al., 2012). The variation in customers' demand makes it difficult for food retailers to forecast accurately what will be required and what to be ordered. This variation is caused by internal (e.g. promotion, price reduction, stock outs) and external factors (e.g. weather, holidays) (Van Donselaar et al., 2006). These factors produce two types of errors (Agnew and Thornes, 1995): overstocking errors and understocking errors. Especially in food retail sector, the overstocking errors in sales forecasting lead to inadequate shelf space, shrinkage, price reductions and food wastage. The understocking errors result in lost sales, lower customer confidence and deterioration of market image. Therefore, it is important to consider these influencing factors in forecasting model to improve forecast accuracy. The eventual purpose associated with a forecasting model is to place an order based on the forecast's outcome. Although the forecasting model is good, there exists always an uncertainty related to the forecast which affects the inventory management decisions (Fleisch and Tellkamp, 2005). To overcome this issue, it is necessary to use quantile predictions, instead of a point forecast or extrapolated quantiles. Bearing these issues in mind, this study focuses on developing a time series model with external variables to forecast daily sales of a perishable food using a hybrid SARIMA-QR model. The rest of this paper is organized as follows: Section 2 describes briefly the current state-of-art of food sales forecasting and demand influencing factors. Section 3 presents the development of SARIMAX models using MLR and QR. Section 4 presents the description of data. Section 5 explains the implementation of the developed model. The results are discussed in Section 6. Finally, Section 7 presents the conclusion.

2. Literature review

In the past several decades, most of the researchers in the field of sales forecasting proposed new forecasting techniques, evaluated the performance of existing ones or modified the existing ones depending on applications (Zotteri et al., 2005). The techniques ranges from simple moving average to advanced machine learning system. But in the area of food sales forecasting, the recent publications are very less when compared to other fields.

2.1. Food sales forecasting

Kong and Martin (1995) applied back propagation neural network (BPNN) to forecast future sales volumes of a food product for a large Victorian food wholesaler. They also suggested from the results that the poor selection of parameter settings in BPNN model can lead to slow convergence and/or incorrect output. To tackle the uncertainty in seasonality, Chang (1997) presented a fuzzy forecasting technique for the seasonality in time series food sales. He analyzed both seasonal and trend fuzziness in his study. Doganis et al. (2006) presented a nonlinear time series sales forecasting model, which is a combination of radial basis function neural network and genetic algorithm to forecast the sales of fresh milk in a major manufacturing company of dairy product. They also proved that the adaptive neural network gives less forecast error when compared to other linear and nonlinear time series models. But the selection of parameters for genetic algorithm and number of nodes in neural network makes this model complex. Taylor (2007) developed an exponentially weighted quantile regression method, which generates interval forecasts from

quantile predictions. He also recommended the application of interval forecasting method when the time series are highly volatile and skewed due to extreme values. His method provides better results when compared to traditional methods. Chen and Ou (2009) developed a model which integrates gray relation analysis and multi-layer functional link network to forecast sales of perishable food in a convenience store. They proved this model reduces more percentage of forecast error than other statistical time series (moving average, autoregressive integrated moving average (ARIMA), generalized autoregressive conditional heteroskedasticity (GARCH)) and artificial neural networks (ANN) (BPNN, Generalized BPNN) models. Hasin et al. (2011) suggested a fuzzy ANN approach to forecast sales of selected products (including perishable foods) in a retail chain in Bangladesh. They also proved that the forecasting performance of fuzzy ANN is better than holt-winter's model in terms of mean absolute percentage error (MAPE). Lee et al. (2012) used a sales forecasting model for convenience stores using BPNN and compared the results with moving average and logistic regression. Shukla and Jharkharia (2013) applied ARIMA model to forecast the demand of vegetable on daily basis in an Indian wholesale vegetable market with a MAPE in the range of 30%. Žliobaitė et al. (2012) presented a two level switch model and studied a case of food wholesaler Sligro Food Group N. V. This sales prediction approach divides the time series sales into predictable and random, then uses intelligent predictor for predictable and moving average for random.

2.2. Hybrid models

There is no universal forecasting model which can be applicable for different kinds of problems. For example, it cannot be expected that a model which predicts price of a product with better accuracy, may provide good results in predicting demand of the same product. Therefore, the forecast accuracy can be improved only if two or more models with different capabilities are integrated and applied rather than a single specific model with limited capability.

Bratina and Faganel (2008) developed an Auto Regressive Moving Average with external variables (ARMAX) model to forecast daily beer demand in Slovenian market. Cools et al. (2009) used both ARIMAX and SARIMAX models to forecast the daily traffic counts. Lee and Hamzah (2010) developed an ARIMAX model to forecast monthly sales of Muslim boys' clothes in Indonesia. This model combines ARIMA model and calendar variation effect during Eid holidays using linear regression and had better forecast results than decomposition method, SARIMA, and ANN. Chikobvu and Sigauke (2012) predicted the daily peak electricity demand in South Africa using SARIMA and regression-SARIMA. Cornelsen and Normand (2012) used an ARIMAX model to estimate the impact of smoking ban on bars in Ireland. Peter and Silvia (2012) identified the advantage of ARIMAX model over ARIMA model in their study to forecast quarterly gross domestic product (GDP) per capita. In order to reduce mortality of silver eel during migration and to optimize shutdowns of turbines, Trancart et al. (2013) applied the SARIMAX model using the fish migration data from two fishing sites in Brittany (north-western France). Kongcharoen and Kuangpradit (2013) used ARIMAX model to forecast monthly exports from Thailand to its trading partners using composite leading indicator as external variable.

In addition to hybrid (S)ARIMA- linear regression models, many of the past studies also developed hybrid (S)ARIMA-ANN models. Luxhøj et al. (1996) developed one of the earliest hybrid econometric-ANN models and applied to forecast the total monthly sales of consumer goods for a Danish company. In their model, they attempted to integrate a seasonally adjusted time series component, an econometric model with lagged external

variables and an ANN component. Zhang (2003) proposed a hybrid ARIMA-ANN model, which outperformed the individual models for Wolf's sunspot data, Canadian lynx data and exchange rate time series data. Another hybrid SARIMA and ANN was presented by Aburto and Weber (2007) to forecast daily demand in a Chilean supermarket. Their proposed model outperforms existing naïve, seasonal naïve, unconditional average, SARIMAX and several neural network models. Based these studies, many of them tried to apply hybrid ARIMA-ANN model in different fields of application. Maia et al. (2008) presented an interval-valued time series forecasting of meteorological and stock price data using a hybrid ARIMA-ANN model and compared it with autoregression (AR), ARIMA and ANN models. Díaz-Robles et al. (2008) introduced a hybrid ARIMA-ANN model to forecast air quality in Chile. Cadenas and Rivera (2010) recommended a hybrid ARIMA-ANN model to forecast wind speed. Shukur and Lee (2015) also suggested the use of hybrid kalman filter and artificial neural network (KF-ANN) model to improve the accuracy of wind speed forecasting. Jeong et al. (2014) proposed a hybrid SARIMA-ANN model, which was shown to provide better accuracy for forecasting annual energy cost budget (AECB) in educational facilities of South Korea than the classical SARIMA model. Babu and Reddy (2014) suggested a hybrid ARIMA-ANN model, which has higher accuracy when compared to the individual models and the existing hybrid ARIMA-ANN models using a simulated data set and experimental data sets such as sunspot data, electricity price data and stock market data.

Likewise, Choi et al. (2011) developed a hybrid forecasting scheme, which combines SARIMA model and wavelet transform (SW) and was shown to perform better than classical SARIMA model, classical seasonal decomposition+linear extrapolation seasonal adjustment method (CSD+LESA) and evolutionary neural networks (ENN) based on forecast accuracy and computational time. Nie et al. (2011) introduced a hybrid model which integrates ARIMA with support vector machines (SVM) to forecast short-term load forecasting for energy management. Aye et al. (2015) compared 26 (23 individual and 3 combination) forecasting models using monthly aggregated retail sales in South Africa. They found that the combination forecasts offer ways of incorporating and discarding information from a larger number of forecasting models. Specifically, the discounted combination forecast model (DISC) outperforms all the individual models, and the other two combination forecast (simple mean and principal component) models. Pektaş and Kerem Cigizoglu (2013) proposed a hybrid ANN-seasonal decomposition model to forecast monthly runoff coefficients to overcome the drawbacks of ARIMA, ARIMAX and ANN. Kriechbaumer et al. (2014) highlighted the use of combined wavelet-ARIMA approach for forecasting monthly prices of aluminum, copper, lead and zinc to improve the forecasting performance of individual models.

2.3. Demand influencing factors

In general, the demand influencing factors reflect the customer behaviors. The food sales forecasting literature found several factors that significantly drive the customers' demand. Aburto and Weber (2007) considered payment, intermediate payment, before holidays, holidays, festivals, school vacation, climate and price variables as input neurons in their hybrid ARIMA-ANN forecasting model. Ali et al. (2009) used type of promotion, price and percentage of discount in their proposed regression tree forecast approach. Ramanathan and Muyldermans (2010) discussed the demand factors influencing the sales of a leading soft drink company in the UK. They considered type of promotion, size of promotion, duration of promotion, price discount, festivals, holidays, temperature, week-in-year, product ranks and new/established as

variables to understand demand structures for different product families using structural equation modeling. They also reviewed some studies on identification of demand influencing factors. Lee and Hamzah (2010) used calendar effect of holidays as external variable in their sales forecasting application. Hasin et al. (2011) recognized holidays (regular holidays, and festivals), promotional activities, availability, price, consumption rate, brand loyalty, and climate as demand influencing factors apart from time series components as input variables in his ANN based forecasting approach. Žliobaitė et al. (2012) used holidays, festivals, school vacation, weather (temperature, pressure, and rain) and promotions as external prediction features in their intelligent model. Poel (2010) and Peters (2012) investigated different demand influencing independent variables which are categorized into location characteristics, promotional variables, weather, national holidays and product characteristics. They analyzed these independent variables in multiple linear regression analysis to predict promotional sales. Sharma and Sharma (2012) considered temperature, day-of-the-week, holidays and sale of alternate products in their ANN approach.

Moreover, some of the literature emphasized the importance of weather in sales forecasting. Agnew and Thornes (1995) explored the weather sensitivity of the food and drinks sector within the UK retail and distribution industry. In their study, the feasibility and benefits of using meteorological knowledge in retail industry were discussed elaborately. They also discussed internal and external forces that trigger the demand. The internal forces are in-store promotions, advertising campaigns, merchandising, price changes, changes in retail outlets and restructuring of management and operational systems. The external forces are economic factors, political and legal considerations, technical development, social trend, seasons, holiday periods and weather variability. Fearn et al. (2006) emphasized the importance of consideration weather in soft fruit sales. Mirasgedis et al. (2014) developed a demand model which incorporates primary meteorological parameters (temperature, precipitation, sunshine duration, and relative humidity), secondary meteorological parameter (apparent temperature), economic variables (tourist activity), time (month-of-the-year and season-of-the-year) with festivals.

Based on the literature review and opinions from the store managers, the demand influencing factors can be classified as shown in Fig. 1. In Fig. 1, the demand influencing factors are classified into events, weather, seasonality, price, substitution and cannibalization, product characteristics and number of customer visits. Out of these factors, the price and the product characteristics are internal forces, which are controllable. But the substitution and cannibalization are partially internal, which cannot be controlled entirely. On the other hand, the other factors are external forces, which are uncontrollable. The events are categorized into regular holidays except festivals, festivals and school vacations. The weather is categorized into air temperature, precipitation, snow cover, sunshine duration, windspeed and relative humidity. The seasonality is categorized into day-of-the-week (weekly seasonality), day of the month (monthly seasonality), month-of-the year (yearly seasonality), yearly seasons and yearly quarters. The price is categorized into normal price and reduced price. The price reduction is classified into promotion (see Section 4.4) and discount (see Section 4.5). The price reduction of the selected product may sometimes produce cannibalization effect on similar type of products in the same store and/or in the competitive stores. The substitution and cannibalization are categorized into promotion, discount or stock-out for similar type of products in the same store and/or in the competitive stores. The promotion for similar type of products (in the same store and/or in the competitive stores) usually produces cannibalization effect on the selected product. The discount for similar type of products (in the same store

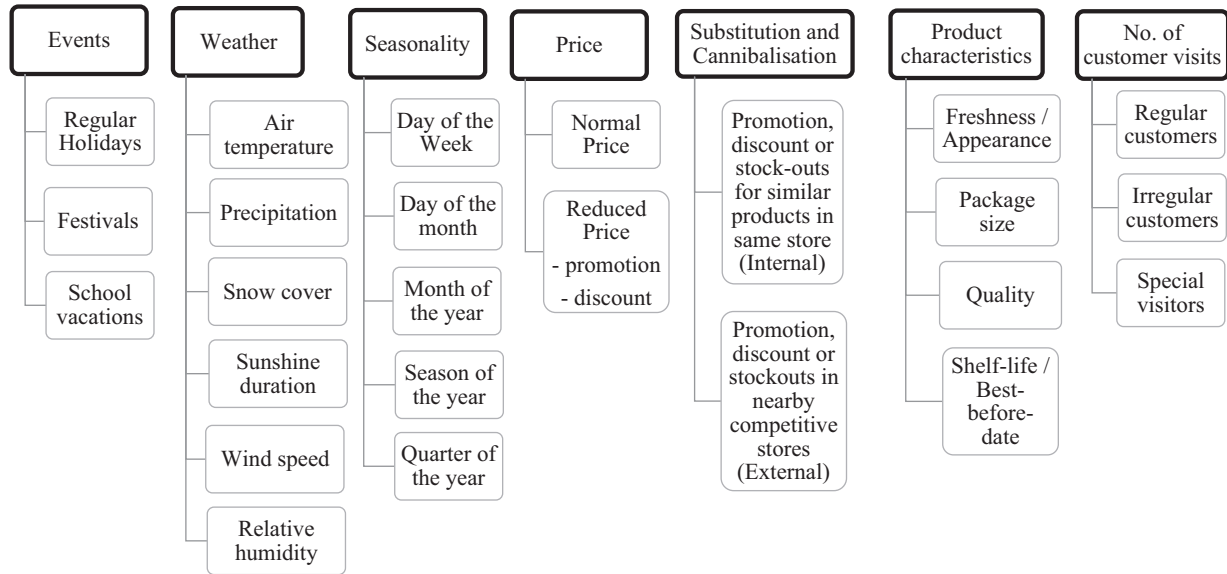


Fig. 1. Classification of demand influencing factors.

and/or in the competitive stores) produces both substitution and cannibalization effects on the selected product. Likewise when there are stock-outs for similar type of products (in the same store and/or in the competitive stores), the selected product acts as a substitute. The product characteristics are classified into product freshness/appearance, package size, quality and shelf-life/best-before-date. The product freshness/appearance depends on the perceptions by the store managers (while discounting or discarding) and the consumers. The product quality implies the physical condition of a product like damage, disease and mold. The shelf-life indicates the life of a product after its arrival in the store. The number of customer visits is categorized into different type of customers visiting the stores such as regular customers, irregular customers and special visitors (e.g. tourists).

3. Methodology

3.1. SARIMA model

An ARIMA model is a statistical technique to predict time series (Box et al., 2008). SARIMA takes into account the seasonal elements in the data with ARIMA modeling. The ARIMA model can be converted to SARIMA model by including seasonal autoregressive, seasonal moving average and seasonal differencing operators. If the time series X_t is modeled as SARIMA $(p, d, q) \times (P, D, Q)_S$ process, it can be stated as (Cools et al., 2009; Aburto and Weber, 2007):

$$\varphi_p(B)\Phi_P(B^S)(1-B)^d(1-B^S)^D Y_t = c + \theta_q(B)\Theta_Q(B^S)\epsilon_t \quad (1)$$

where S - Seasonal length (e.g. in quarterly data $s=4$ and in monthly data $s=12$), B - Delay or lag operator a time series observation lag k period is symbolized $B^k X_t = X_{t-k}$, $\varphi_p(B)$ - Autoregressive operator of p -order $(1 - \varphi_1(B) - \varphi_2(B^2) - \dots - \varphi_p(B^p))$, $\theta_q(B)$ - Moving average operator of q -order $(1 - \theta_1(B) - \theta_2(B^2) - \dots - \theta_q(B^q))$, $(1 - B)^d$ - Differencing operator of order d to remove non-seasonal stationarity, $\Phi_P(B)$ - Seasonal autoregressive operator with P -order

$$(1 - \Phi_1(B) - \Phi(B^{2s}) - \dots - \Phi_P(B^{Ps}))$$

$\Theta_Q(B)$ - Seasonal moving average operator with Q -order

$$(1 - \Theta_1(B) - \Theta_2(B^{2s}) - \dots - \Theta_Q(B^{Qs}))$$

$(1 - B^S)^D$ - Differencing operator of order D to remove seasonal stationarity, Y_t - Time series observation at time t , ϵ_t - Residual error, c - Constant

The general SARIMA model consists of four iterative steps:

- (1) Identification: This step involves selection of order of differencing (d), order of seasonal differencing (D), seasonal length (S), non-seasonal autoregressive order (p), seasonal autoregressive order (P), non-seasonal moving average order (q) and seasonal moving average order (Q) using autocorrelations (AC) and partial autocorrelations (PAC).
- (2) Estimation: The historical sales data are used to estimate the parameters of the identified SARIMA model from step 1.
- (3) Diagnosis: The model is diagnosed to check the adequacy until the residuals appear as white noise.
- (4) Forecasting: The diagnosed model from step 3 is used for forecasting the future values.

3.2. SARIMAX model

Instead of modeling a time series Y_t with only a combination of lagged values, can be explained by external variables using linear regression. In linear regression analysis, the errors are assumed to be random in nature. But in case of time series, it is often violated by the autocorrelation property present within the data. This is rectified by SARIMAX, which explains the autocorrelation (Cools et al., 2009). Therefore, the SARIMA is extended into SARIMA model with external variables, called SARIMAX $(p, d, q) \times (P, D, Q)_S X$, where X is the vector of external variables. Using Eq. (1), the general SARIMAX model can be expressed as (Cools et al., 2009; Peter and Silvia, 2012):

$$Y_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \dots + \beta_k X_{k,t} + \frac{\theta_q(B)\Theta_Q(B^S)}{\varphi_p(B)\Phi_P(B^S)(1-B)^d(1-B^S)^D} \epsilon_t \quad (2)$$

where $X_{1,t}, X_{2,t}, \dots, X_{k,t}$ are observations of external variables corresponding to the dependent variable, $\beta_0, \beta_1, \dots, \beta_k$ are regression coefficients of external variables.

3.3. Quantile regression

The point forecasting is a poor decision making aid. The performance of point forecasting reduces when there is an extreme and uncertain situation (Li et al., 2012). Chatfield (2000) mentioned the importance of providing the interval forecasts such as assessing future uncertainty, planning different strategies for the range of possible outcomes, comparing forecasts and exploring different scenarios based on different assumptions. In food retail industry, the estimation of forecast interval is very important for inventory planning to assess the appropriate levels of safety stock. To overcome this issue, the point forecasts were converted into the interval forecasts in most of the earlier studies. Granger et al. (1989) suggested the confidence intervals for 10% and 90% as $(Y_t \pm 1.282\sigma_t)$. The confidence interval of the prediction is a range that is likely to contain the mean response for the given values of independent variables in a model. However, the prediction interval of forecast is a range that is likely to contain the mean forecasted response for the given value of independent variables in a model (Walpole et al., 2012). Chatfield (2000) and Hyndman and Athanassopoulos (2013) mentioned the calculation of prediction intervals using theoretical formulae that are basically of the same general form. Prediction interval at $100(1-\alpha)\%$ for the value h steps ahead can be represented by

$$\widehat{Y}_t \pm z_{\alpha/2} \sqrt{\text{Var}[e_n(h)]} \text{ or } \widehat{Y}_t \pm z_{\alpha/2} \hat{\sigma} \quad (3)$$

where $\hat{\sigma}$ is an estimate of standard deviation of forecast errors $e_n(h)$.

Walpole et al. (2012) recommended another formula

$$\widehat{Y}_t \pm z_{\alpha/2} \hat{\sigma} \sqrt{1 + \frac{1}{n}} \quad (4)$$

These methods can be used, only when the residuals are normal and uncorrelated. Taylor (2007) argued that the Gaussian assumption is unrealistic and shape of the distribution of sales data are often not constant over time. Instead of point forecast based prediction intervals, he suggested the use of quantiles in forecasting. The classical mean regression model describes how mean of response variable (y) changes with vector of covariates (x) based on conditional mean function $E(\frac{y}{x})$. Koenker and Hallock (2001) extended this mean regression model to conditional quantiles of response variable. Unlike mean regression model, QR considers conditional quantile function $Q_p(\frac{y}{x})$. QR approach provides an in-depth sight of the relationship of the dependent variable and the independent variables.

In conditional mean model, the regression coefficients (β) are estimated by minimizing the sum of squared errors (Koenker and Hallock, 2001; Benoit and Van den Poel, 2009):

$$\min_{\beta \in \mathfrak{R}} \sum_{i=1}^n (y_i - x'_i \beta)^2 \quad (5)$$

In conditional median model, the regression coefficients are estimated by minimizing the sum of absolute deviations (Koenker and Hallock, 2001; Benoit and Van den Poel, 2009):

$$\min_{\beta \in \mathfrak{R}} \sum_{i=1}^n |y_i - x'_i \beta| \quad (6)$$

The QR extends the median regression (50%) to all other quantiles and minimizes the objective function for quantile (Koenker and Bassett Jr., 1978):

$$\min_{\beta \in \mathfrak{R}} \sum_{i: y_i \geq x'_i \beta} \tau |y_i - x'_i \beta| + \sum_{i: y_i < x'_i \beta} (\tau - 1) |y_i - x'_i \beta| \quad (7)$$

where $\tau \in (0, 1)$ is any quantile. The quantile β_τ is the τ th regression quantile. When τ is equal to 0.5, it is median regression.

Using the point values in identifying the significance of variables is also one of the drawbacks of classical mean regression model. QR approach allows us to examine the behavior of the response variable beyond the average of the Gaussian distribution. Therefore, QR can be used to build prediction intervals (Meinshausen, 2006). Investigating the behavior of the response variable for different quantiles with respect to the independent variables may result in different inferences in comparison with examining only the average of response variable (Benoit and Van den Poel, 2009).

3.4. Cross-validation

There is always a need to trade-off between the amounts of resources invested on forecasting to achieve a selected accuracy and the amounts of paybacks from them. According to Gilliland and Sglavo (2010), the reasonable forecasting performance objective is to beat a benchmark model and to improve the forecasting process continuously. They suggested a better practice is to use a simple naïve forecasting model as a benchmark to scale the forecasting performance objective. The forecast accuracy of the naïve model can act as a baseline against the proposed model. If the proposed model does not perform up to the benchmark model, then it can be substituted by the benchmark model itself. In addition to this, Gilliland (2011) suggested the use of forecast value added (FVA) analysis to compare the forecasting methods. According to him, FVA analysis can be used to identify and eliminate those activities that failing to improve the forecast, while simultaneously yielding better forecast results.

Cross-validation is primarily a way of measuring the forecasting performance of a model. The usual way to cross-validate a model is to test it on a set of data not used in fitting. In order to measure the forecast accuracy of a model, mean errors of the test data set can be used. This study uses MAPE and root mean squared error (RMSE) as performance measures. The MAPE is defined as

$$\frac{1}{n} \sum_{t=1}^n \left| \frac{Y_t - \widehat{Y}_t}{Y_t} \right| \quad (8)$$

and the RMSE is defined as

$$\sqrt{\frac{\sum_{t=1}^n (Y_t - \widehat{Y}_t)^2}{n}} \quad (9)$$

where $\widehat{Y}_t$ is the forecast and n is the number of observations.

4. Description of data

This study uses daily sales data of banana measured in kilograms from a typical food retail store as a case study from January 2010 to April 2014. The selected food retail store is in Lower Bavaria (Germany), close to upper Austrian border. The banana is selected a perishable food for this study because of its short shelf-life (2–4 days) and its availability throughout the year. Fig. 2 illustrates the daily sales data of banana for 2013 in time series plot. The mean sales per day is about 64 kg (Kg). This time series is highly periodic, but it is difficult to observe the seasonal patterns. The seasonal variability can be investigated by analyzing the impact of day-of-the-week and month-of-the year seasonalities on daily sales data. In addition to these effects, there are holiday effect, promotional effect, discount effect and weather effect. As mentioned in Section 2.2, this study attempts to assimilate all the effects of the demand influencing factors. Nonetheless, the effects

due to the substitutions, the product characteristics and the customer visits are not taken into account due to lack of data and information. The description and representation of the considered effects are given as follows.

4.1. Day effect

As the store is closed on Sundays, Week days from Monday to Saturday are used in calculation. In Fig. 3, the box-and-whisker plot displays the median daily sales from Monday to Saturday. From this figure, it is evident that the highest sales usually occurs on Saturday. Friday and Monday are the next highest sales days. The box-and-whisker plot also shows information about extreme values (o) and outliers (*), which occur mainly due to holidays and promotions. However, the median of daily sales for Friday and Saturday are similar, Friday has more dispersion for sales than Saturday. The spectral analysis is a statistical approach to discover regular cyclical patterns or periodicities. In spectral analysis, the sales data are transformed with fourier transformation and decomposed into waves of different frequencies (Coshall, 2000). Fig. 4 shows the spectral density associated with daily sales of banana against frequencies. From this figure, the spectral density reaches a global maximum around frequency 0.165 ($\sim 1/6$) and a

second maximum around frequency 0.33. This global maximum can be interpreted as a weekly periodicity.

4.2. Month effect

Usually, the average sales of banana during summer months is higher than other seasons. This type of month effects can be recognized by considering the months from February to December as dummy variables (0 and 1). The reference month is January, so coefficients of other months may be interpreted relative to this month.

4.3. Holiday effect

In this study, the holidays are classified into four categories such as regular holidays, festivals, upper Austrian holidays (except German holidays) and school vacations. The regular holidays include the official nine holidays in Bavaria, Germany except Easter and Christmas. The festivals include Easter and Christmas. Usually, the stores are closed during regular holidays and festivals. The missing sales data during regular holidays and festivals are filled with the estimated values from trend using linear interpolation, in order to maintain the alignment of data between time periods, i.e., seasonality. The upper Austrian region has extra three holidays than Bavarian regular holidays. The upper Austrian holidays are considered only because of cross-border shopping. To evaluate the holiday effects, the effects are categorized into before holidays, after holidays, Christmas (before and after), Easter (before and after), before and on Upper Austrian holidays and school vacations. The holiday effects can be incorporated into the model using dummy variables, i.e., ordinary days are coded as 0 and days with effects are coded as 1.

4.4. Promotional effect

In this study, the promotion of banana is of two types: planned and unplanned. The planned promotion is a price reduction that appears in advertisement sheet or above product display. This promotion exists for complete week. The unplanned promotion is a price reduction that is done unexpectedly on any days in a week based on the price of a product from competitors. This promotion usually exists for two or three days. However, the promotional effect in this model is used as a single variable and is expressed in percentage of price reduction.

4.5. Discount effect

Discount is an unplanned price reduction that is done by the store manager on banana after its two or three days of shelf-life due to decrease in quality. Although the discounted sales of banana helps in food waste reduction, it creates disturbances (noise) in the regular sales data which in turn reduces the forecast accuracy. The weighted average discount sales in a day is estimated by sum of the products of discount sales and their corresponding percentages of discount, divided by sum of percentages of discount. Here the percentage of discount acts as weightage.

$$\text{Weighted average discount sales} = \frac{\sum_{i=1}^m S_i \cdot D_i}{\sum_{i=1}^m D_i} \quad (10)$$

where m – Number of discounts in a day, D_i – Percentage of i^{th} discount, S_i – Discount sales for corresponding D_i

4.6. Weather effect

In case of food retail industry, the extreme weather conditions play a minor role in sales increase and decrease (Agnew and

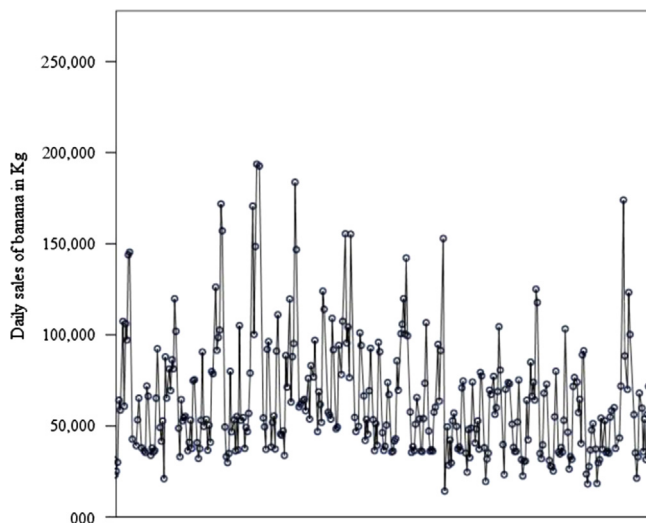


Fig. 2. Daily sales of banana for a retail store during 2013.

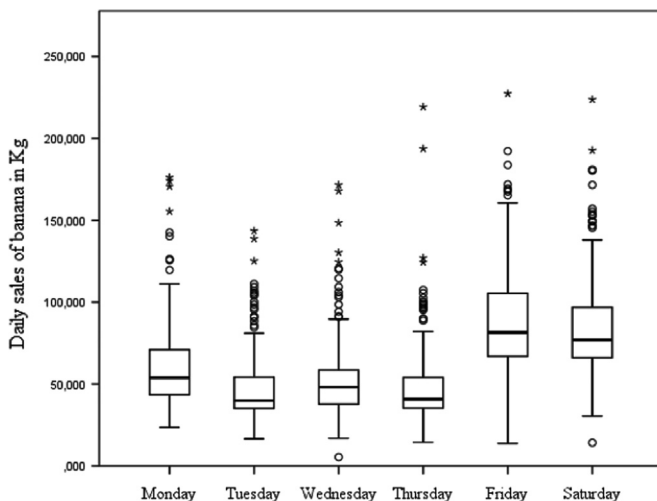


Fig. 3. Box-and-whisker plot for daily sales of banana from Monday to Saturday.

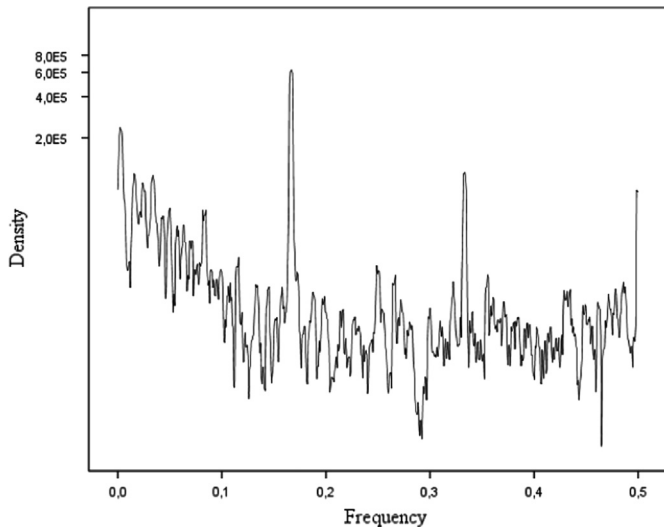


Fig. 4. Spectral density associated with daily sales of banana against frequencies.

Thornes, 1995). In this study, the weather parameters such as maximum air temperature (degrees Centigrade), precipitation (mm), fresh snow depth (cm), snow depth (cm), sunshine duration (hours) and relative humidity (%) are considered and the data are collected from German Weather Service (Deutscher Wetterdienst (DWD)) (from January 2010 to April 2014). The weather station is situated nearly 5 km from the food retail store under study. The collected weather data are used here to model the extreme weather conditions, not the seasonal effects, which can be modeled using month-of-the year seasonality. Maximum air temperature, precipitation, sunshine duration and relative humidity of a day are the average values from 7 am to 8 pm (around working hours of store) at the selected weather station. The fresh snow depth is the snow depth in last 24 h. The fresh snow depth and snow depth are taken as the average values of a day.

5. Application of SARIMAX models

The SARIMAX modeling can be carried out in two steps: (i) developing a SARIMA model and (ii) the predicted values of SARIMA model are included as an input variable along with external variables using MLR and QR. Then, the diagnosis and parameter significance testing are performed to check the residuals are white noises.

5.1. Developing a SARIMA model

As mentioned above, the initial step of the SARIMAX is to develop a SARIMA model for the given data. For the purpose of cross-validation, the entire dataset is divided into two sets: training set containing 70% of the observations and validation set containing 30% of the observations. The autocorrelation and partial autocorrelation plots of daily sales of banana for 16 days are shown in Fig. 5a and 5b respectively. The ACF of the training set shows that there is no regular or seasonal trend. Augmented Dickey–Fuller test (24 lags) also proves that the time series is stationary. So, regular or seasonal differencing is not required. The autocorrelation and partial autocorrelation were investigated in training set, to identify the orders of SARIMA model. SARIMA (0,0,4) (1,0,1) (6) model is identified as the best model from the ARIMA family as mentioned in Section 3.1. The results of SARIMA model are obtained using *forecast* package in R (Hyndman et al., 2014) and SPSS 22. Using Eq. (2), the estimates of this final SARIMA

model can be represented by the following equation:

$$\begin{aligned} (1 + 0.999B^6) Y_t = & 67.01 + (1 + 0.3197B + 0.3394B^2 + 0.2020B^3 \\ & + 0.1227B^4) (1 - 0.9652B^6) \varepsilon_t \end{aligned} \quad (11)$$

The performance of the SARIMA model was measured in terms of MAPE and RMSE for in-sample (training) and out-sample (test) data, are shown in Table 1. The results of the SARIMA model show that the model's performance is consistent both in training and testing. The residuals appear to be not normally distributed. Ljung–Box Q test also proves the same.

5.2. SARIMA-MLR model

The second step of the methodology is to fit a MLR model with ordinary least squares and simultaneous entry of all independent variables, where the daily sales of banana (Y_t) is regarded as the response variable and the predicted sales using SARIMA model along with other external variables as the independent variables. The external variables are Christmas, Easter, school vacation, before holiday, after holiday, Austrian holiday, February, March, April, May, June, July, August, September, October, November, December, discounted sales, promotion, relative humidity, temperature, sunshine duration, fresh snow depth and snow depth. The day-of-the-week effects were not included in the SARIMAX

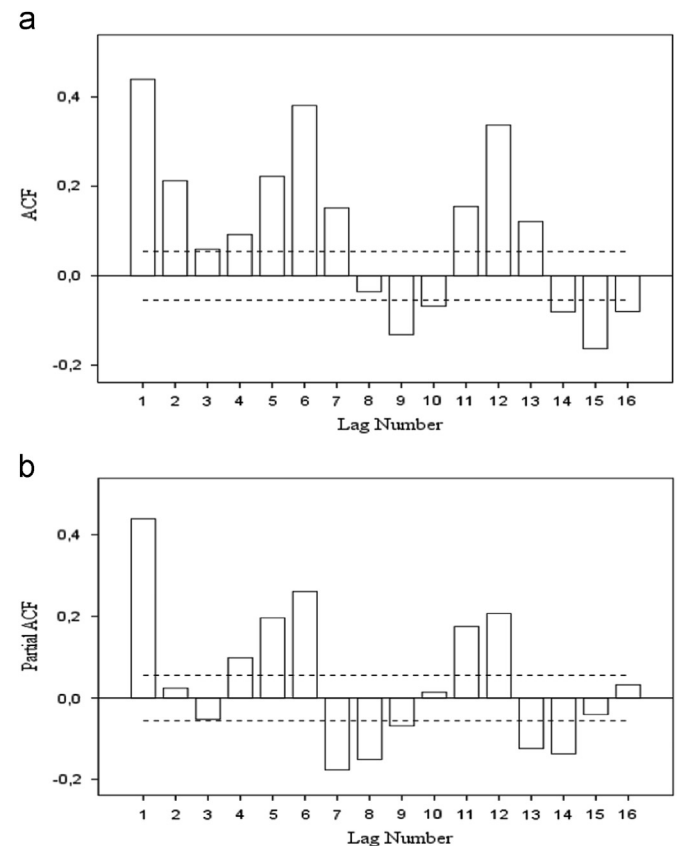


Fig. 5. (a) Autocorrelation and (b) partial autocorrelation plot of sales of banana.

Table 1

Performance of SARIMA for training and test data set of daily sales of banana.

Performance measures	Training	Test
MAPE	27.57	28.78
RMSE	23.37	22.03

Table 2
Overview of variables in the analysis.

Name of variable	Notation	Unit
Dependent variable		
Daily sales of banana	Y	Kilogram ^a
Independent variables		
Predicted daily sales of banana (using SARIMA)	x1	Kilogram ^a
Christmas	x2	1 or 0 ^b
Easter	x3	1 or 0 ^b
School vacation	x4	1 or 0 ^b
Before holiday	x5	1 or 0 ^b
Austrian holiday	x6	1 or 0 ^b
After holiday	x7	1 or 0 ^b
February	x8	1 or 0 ^b
March	x9	1 or 0 ^b
April	x10	1 or 0 ^b
May	x11	1 or 0 ^b
June	x12	1 or 0 ^b
July	x13	1 or 0 ^b
August	x14	1 or 0 ^b
September	x15	1 or 0 ^b
October	x16	1 or 0 ^b
November	x17	1 or 0 ^b
December	x18	1 or 0 ^b
Discounted sales	x19	Kilogram ^a
Promotion	x20	Percentage ^a
Precipitation	x21	Millimeter ^a
Sunshine duration	x22	Hours ^a
Freshsnow depth	x23	Centimeter ^a
Snow depth	x24	Centimeter ^a

^a Continuous variable.

^b Dummy variable.

Table 3
Combined SARIMA-MLR model for daily sales of banana.

	Estimate	Std.error	t	p-value	VIF ^a
Intercept	4.3848	2.5047	1.7510	0.0802	–
Predicted daily sales of banana (using SARIMA)	0.8183	0.0266	30.7780	< 0.0001	1.1552
Christmas	8.1244	5.6295	1.4430	0.1492	1.1701
Easter	41.7502	5.2945	7.8860	< 0.0001	1.1811
School vacation	0.4864	1.6557	0.2940	0.7690	1.7150
Before holiday	24.2388	3.2383	7.4850	0.0000	1.0845
Austrian holiday	–7.7530	4.1538	–1.8660	0.0622	1.0838
After holiday	1.2581	3.3271	0.3780	0.7054	1.0892
February	–0.3643	2.5966	–0.1400	0.8884	1.9771
March	4.2469	2.4764	1.7150	0.0866	1.9571
April	2.3300	2.7620	0.8440	0.3991	2.0182
May	6.7913	2.7179	2.4990	0.0126	1.9212
June	7.6321	2.7443	2.7810	0.0055	1.9079
July	2.3492	2.7386	0.8580	0.3912	1.9505
August	3.0049	3.0776	0.9760	0.3291	2.4845
September	–3.7452	2.7490	–1.3620	0.1733	1.8973
October	–2.4092	2.6868	–0.8970	0.3701	1.8774
November	–3.7272	2.6442	–1.4100	0.1589	1.7870
December	–5.8502	2.6896	–2.1750	0.0298	1.8813
Discounted sales	0.4724	0.1830	2.5820	0.0099	1.0522
Promotion	1.1677	0.0645	18.0990	< 0.0001	1.1318
Precipitation	–0.0674	0.1570	–0.4290	0.6678	1.1787
Sunshinehours	–0.3212	0.1466	–2.1920	0.0286	1.3993
Freshsnow depth	–0.3238	0.2793	–1.1600	0.2465	1.2953
Snow depth	0.0910	0.0489	1.8610	0.0630	2.0488

^a Variance inflation factor.

model, since the seasonal AR and MA parameters were considered in the SARIMA model. Prior to running of the regression analysis, variance inflation factor (VIF) is used as a diagnostic tool to examine the impact of multi-collinearity in this MLR model. The variance inflation factors greater than 5 were used to identify problematic relationships among variables. As a result, the variables representing relative humidity and temperature were excluded from the MLR model. The selected variables with their notations and units are listed in Table 2. The results of the MLR model shows that a total of nine variables-predicted sales using SARIMA, Easter, before holiday, May, June, December, discounted sales, promotion, and sunshine duration-were found to be significant at 5% level. The independent variables such as Austrian holiday, March and snow depth were found significant at 10% level. The adjusted coefficient of determination (adjusted R^2) is 0.62. The results of the regression analysis are presented in Table 3. The intercept of this model can be interpreted as the estimated function of the daily sales of banana during January (neither Christmas nor Easter), not before holidays, not before Austrian holidays, not after holidays and with mean percentage of promotion, mean discount sales, mean precipitation, mean sunshine duration, mean fresh snow depth, and mean snow depth (Koenker and Hallock, 2001). The variables representing predicted sales using SARIMA, Easter, before holiday, March, May, June, discounted sales, promotion, and snow depth have positive effect on the sales of banana, while the variables representing Austrian holidays, December, and sunshine duration have negative effect. The decrease in the sales of banana during Austrian holidays may be due to the absence of Austrian customers during their holidays. The MAPE and RMSE of this SARIMAX model for out-sample data set are 24.18 and 19.14, shows that the SARIMAX model outperforms SARIMA model.

5.3. SARIMA-QR model

In conventional least square regression method which was used in SARIMAX, it has been recognized that the resulting estimates of different covariates on the conditional mean of sales were not reflecting the real size and nature of these covariate effects on the lower and upper tail of the sales distribution. Only quantile functions can provide a thorough depiction of the covariate effects (Benoit and Van den Poel, 2009). Therefore, QR is applied to regress the response variable with covariates. In Fig. 6, the sales of banana in Kg are plotted against quantiles from 0.05 to 0.95 for all independent variables and an intercept. The coefficients and standard errors of all covariates for all the considered quantiles are shown in Appendix. The plots and results of QR are estimated using *quantreg* package in R (Koenker, 2006). The solid line in black color with filled dots represents the regression estimates for quantiles ranging from 0.05 to 0.95. The shaded gray area depicts a 90% pointwise confidence interval for the quartile regression estimates. The red line shows the regression estimate of the mean effect. The two dotted red lines indicate 90% confidence intervals for the least square estimate.

5.4. Cross validation and comparison

In the retail store, which is under study, the store manager follows a seasonal naïve forecasting model to place orders. The seasonal naïve forecasting model forecasts the daily demand based on the previous week sales on the same day-of-the-week. Whenever there are promotions and/or holidays in the previous week, the additional sales due to promotions and holidays are discarded from the corresponding forecasts. Likewise when there are promotions and/or holidays in the forecasting week, the additional sales due to promotions and holidays are added to the corresponding forecasts. With reference to Section 3.4, the seasonal naïve forecasting model is going to act as a benchmark model to represent a baseline

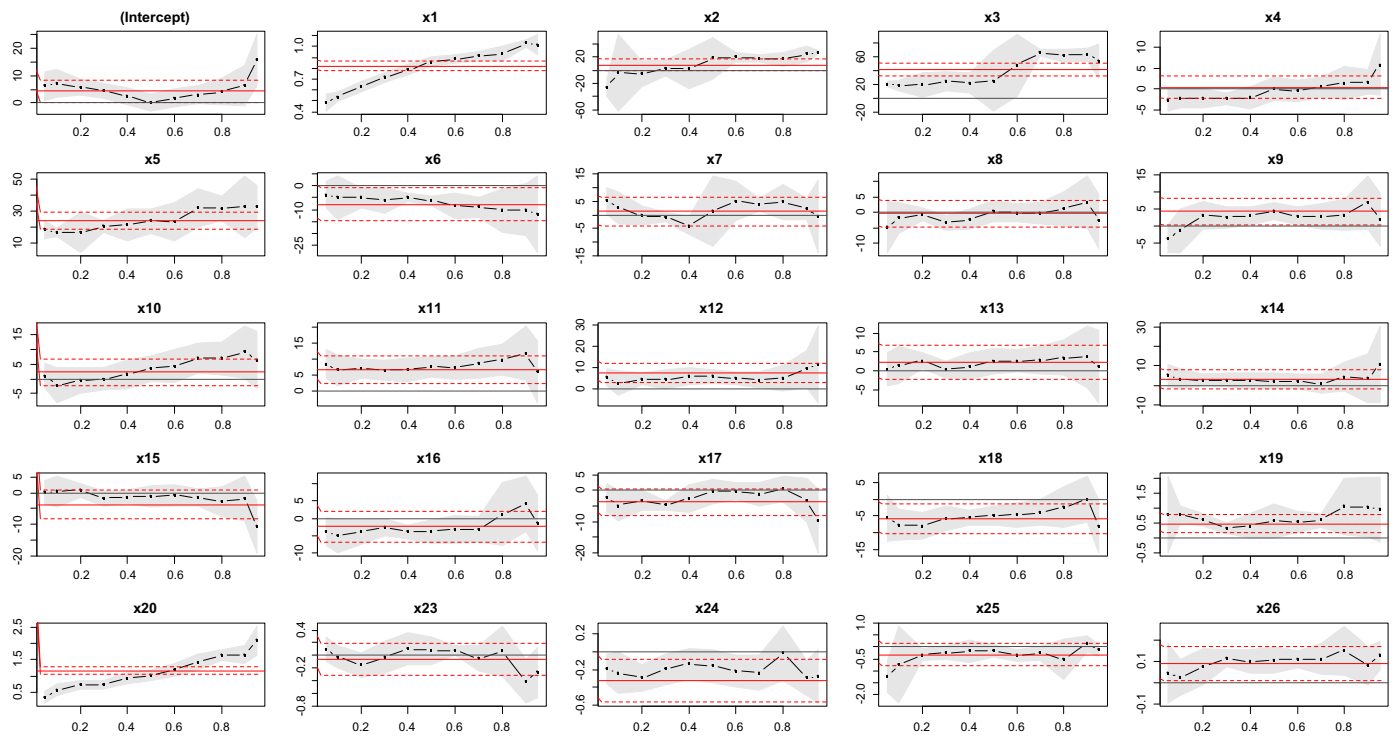


Fig. 6. Ordinary least square and quantile regression plots. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

forecasting model for comparison. In addition to the seasonal naïve forecasting model, the proposed hybrid model is also compared with the individual models such as SARIMA and MLPNN models. The MLPNN is an universal function approximator, which is one of the significant advantage over other non-linear models (Maia et al., 2008). The dependent variable and covariates are rescaled using normalization to improve the network training. The data is partitioned into training (70%) and test (30%) datasets. In this study, the sigmoid function is used as activation function in the hidden as well as the output layers. This function can be expressed as $g(x) = \frac{1}{1+e^{-x}}$ which transforms the real-value arguments to the range (0, 1). The MLPNN is trained using a gradient descent algorithm in this study, adjusting the weights towards convergence using the gradient (Haykin, 1998). Initial learning rate and momentum are taken as 0.4 and 0.9 respectively. In this study, two MLPNN models are used:

MLPNN-1: Three layers with 37 input neurons (holidays, day-of-the-week, month-of-the-year, discount, promotion, and six lags), 9 hidden neurons and one output neuron.

MLPNN-2: Four layers with 37 input neurons, 12 and 9 hidden neurons in two layers and one output neuron.

The MLPNN models in this study are run using neural network module in SPSS 22. The MAPE and RMSE for test dataset using all the models are listed in Table 4.

Obviously, the performance of SARIMA is better than the seasonal naïve forecasting model. But, its performance is poor when compared to MLPNN and SARIMAX models. Although the MLPNN model considers all the demand influencing factors, its forecast accuracy is less than SARIMAX models. As highlighted by Dorfner (1996) in his study, this may be due to some of the important disadvantages of neural network models such as limited availability of sample data, over-fitting, and inclusion of linear case in a trivial way. The SARIMA-MLR and SARIMA-QR models have better forecast accuracy than the other models due to their integration of seasonal adjustment using SARIMA and consideration of external variables using MLR and QR respectively. The practical advantage

Table 4

MAPE and RMSE for test dataset of daily sales of banana using all the models.

Forecasting method	MAPE	RMSE
Seasonal naïve forecasting	35.28	23.08
SARIMA	28.78	22.03
MLPNN-1	27.97	19.89
MLPNN-2	26.53	18.89
SARIMA-MLR	24.18	19.14
SARIMA-QR (Median)	23.47	19.35

of SARIMA-QR model over SARIMA-MLR model is explained in results and discussion (Section 6).

5.5. FVA analysis

As mentioned in Section 3.4, the FVA analysis begins with the assumption that the seasonal naïve forecasting model is not having any effect on the forecasting performance. The FVA analysis can be done using the following steps:

- (1) Calculation of forecast accuracy using MAPE (calculated in Table 4) for different models.

$$\text{Forecast accuracy} = (100 - \text{MAPE})\% \quad (12)$$

- (2) Estimation of FVA (% points), which is the difference between forecast accuracies of existing/proposed model and benchmark model.

$$\text{FVA} = (\text{Forecast accuracy of existing/proposed model} - \text{Forecast accuracy of benchmark model}) \quad (13)$$

- (3) Estimation of comparative improvements in forecast accuracy produced by different forecasting models.

Table 5
FVA analysis.

Forecasting method	Forecast accuracy (%)	FVA over seasonal naïve forecasting (% points)	Comparative improvement in forecast accuracy (% points)
Seasonal naïve forecasting	64.72	–	–
SARIMA	71.22	6.5	6.5
MLPNN-1	72.03	7.31	0.81
MLPNN-2	73.47	8.75	1.44
SARIMA-MLR	75.82	11.1	2.35
SARIMA-QR (Median)	76.53	11.81	0.71

Based these steps, the FVA analysis is conducted and the results are presented in Table 5. The seasonal naïve forecasting model has 65% accuracy. As shown in column 3 of Table 5, the SARIMA, MLPNN-1, MLPNN-2, SARIMA-MLR and SARIMA-QR models improve the forecast accuracy 6.5, 7.31, 8.75, 11.1, and 11.81 percentage points over the seasonal naïve forecasting model respectively. In column 4 of Table 5, the comparative improvements in forecasting accuracy by different forecasting models are presented. MLPNN-1 and -2 are adding value by improving the forecast accuracy 0.81 and 2.25 percentage points against SARIMA model respectively. The hybrid models SARIMA-MLR and -QR are adding value by improving the forecast accuracy 2.35 and 3.06 percentage points against MLPNN-2 respectively. From this analysis, it has been identified that SARIMA-MLR and SARIMA-QR perform better than other individual models.

6. Results and discussion

6.1. Interpretation of SARIMA-MLR and -QR models

In conventional least square regression method, which was used in SARIMA-MLR model, it has been recognized that the resulting estimates of different covariates on the conditional mean of sales were not reflecting the real size and nature of these covariate effects on the lower and upper tail of the sales distribution. In Appendix and Fig. 6, the intercept (c) is different from zero in MLR and is significant from 0.05 to 0.3 quantiles and for 0.95 quantile in QR. The predicted sales of banana (using SARIMA) is significant in all quantiles. It clearly has a positive impact on the expected sales of banana. The estimated conditional quantile function reveals that the effect of the covariate is getting more extreme when the quantiles are higher. The effect on conditional mean and conditional median are similar. In case of Christmas effect, the upper quantile estimates are significant and have a positive effect on the sales of banana. For 0.05 quantile, Christmas has a negative coefficient. This may be due to stock-outs. However, the ordinary least square (OLS) estimate suggests that the effect of Christmas variable is not different from zero. This is a salient example of how misleading the ordinary least square estimates can be. Easter effect variable has an increasing positive effect from lower quantiles to upper quantiles. It is almost significant for all quantiles except 0.5 quantile. The OLS regression strongly overestimates the effect of Easter variable than the median quantile estimate. The school vacation estimate is not different from zero for any quantiles except 0.3 and 0.05 quantiles and has negative effect on the sales. The relationship between the sales and before

holiday variable is positive. The effect of before holiday is significant for all quantiles and increases from lower quantiles to upper quantiles. The Austrian holiday variable has a negative effect and is significant from 0.2 to 0.8 quantiles. It has a decreasing effect from lower quantiles to upper quantiles. The after holiday variable is not different from zero in most of the quantiles except 0.05, 0.4, and 0.8 quantiles.

According to OLS regression, only the month covariates such as May, June and December are significant. The OLS estimate for other months are not different from zero. The February month variable is significant only for 0.3 quantile and has a negative coefficient. The March month variable is significant only in middle quantiles and has a positive effect. For the March variable, the coefficients of conditional mean is similar to the coefficient of conditional median of the sales distribution. The April month variable is significant from 0.7 to 0.9 quantiles and has a positive effect. The May month variable is significant for all quantiles except 0.95 quantile and has a positive effect. The OLS regression overestimates the effect of this variable than the median quantile estimate. The June month variable is significant in most of the quantiles except 0.1, 0.3, 0.8, and 0.95 quantiles and has a positive effect. The July month is significant only for 0.2 quantile and has a positive effect. The August month variable is not different from zero, both in OLS and QR. The September month variable is significant only for 0.95 quantile and has a negative effect. The October month variable is significant from 0.1 to 0.5 quantiles and has a negative effect. The November month variable is significant only for 0.3 quantile and has a negative effect. The December month variable is significant from 0.1 to 0.7 quantiles and has a negative effect. The discounted sales variable is significant for lower (0.1–0.3) and upper (0.6–0.9) quantiles, and has a positive effect. The promotion variable is highly significant and has a positive relationship with the sales of banana for all quantiles. The estimates for promotion increases from lower quantiles to upper quantiles. The precipitation covariate is not different from zero in OLS regression. In QR, it has a significant negative estimate for 0.9 quantile. The sunshine duration variable is significant from 0.05 to 0.95 quantiles and has a negative effect. The fresh snow depth has significant negative estimates for 0.05, 0.2, and 0.6 quantiles. Conversely, the fresh snow depth is not significant in OLS regression. The snow depth is significant from 0.3 to 0.8 quantiles and for 0.95 quantile. The estimates of snow depth variables in QR as well as OLS regression are positive. Theoretically, snow depth has both positive and negative impact on retail food sales. Agnew and Thornes (1995) also suggested that the adverse weather usually have negative effect on the larger food retail stores (supermarkets) located outside the cities, in contrast, positive effect on the neighborhood retail stores. The positive coefficient of snow depth suggests that the retail store under study has a favorable effect when there is an increase in snow depth.

6.2. Prediction interval

The QR is not only improving the interpretation of results from the SARIMA-QR model, but also provides better prediction intervals. The estimators of different quantiles can provide information to management for inventory planning to target a specific service level of sales. The upper and lower bound of the prediction interval delivers optimistic and pessimistic predictions of the sales of banana (Benoit and Van den Poel, 2009). A 95% prediction interval for the value of Y can be given by quantiles $[Q_{0.025}(X), Q_{0.975}(X)]$ (Meinshausen, 2006). In Fig. 7, the observed in-sample values of sales of banana are plotted along with their

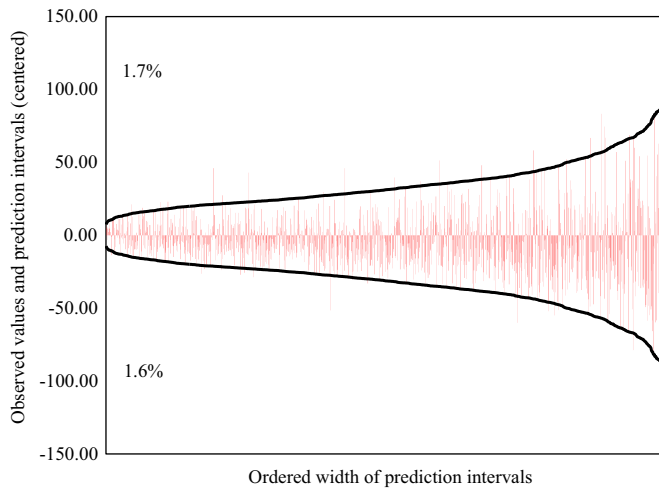


Fig. 7. Graphical representation of 95% prediction intervals from SARIMA-QR for in-sample data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

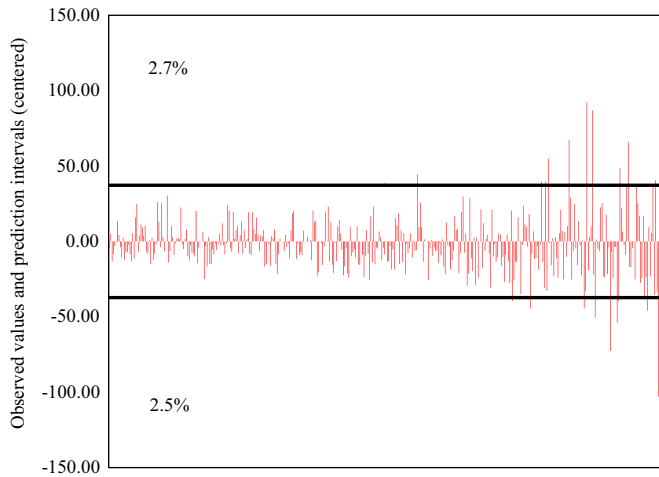


Fig. 8. Graphical representation of 95% prediction intervals from SARIMA-MLR for out-sample data.

prediction intervals based on the graphical representation suggested by Meinshausen (2006). The red columns indicate the difference between the mid-point of the prediction intervals and all the observed values. The black lines indicate the difference between the mid-point and ends of the prediction intervals. These values are plotted against the ordered width of the prediction interval, i.e., the difference between the upper and lower end of the prediction interval. Theoretically, about 95% of all observations are expected to be within 95% prediction intervals. About 5% of all observations should fall outside their prediction intervals. As shown in Fig. 7, all except 45 observations lie inside their corresponding prediction intervals. The percentage of the observed values that lie above the upper prediction interval is specified in the upper left corner (1.7%). The percentage of the observed values that lie below the lower prediction interval is specified in the lower left corner (1.6%). This same procedure is used for the out-sample data to compare the prediction results SARIMA-QR and SARIMA-MLR models. The prediction intervals (5% and 95%) are estimated from the point forecasts of SARIMA-MLR model using Eq. (3) suggested by Chatfield (2000) and Hyndman and Athanasopoulos (2013). The prediction intervals estimated from SARIMA-

MLR model are plotted against the observed out-sample values of sales of banana in Fig. 8. As shown in this figure, ~5.5% of the observed values lie outside the prediction intervals. Likewise, Fig. 9 shows the prediction intervals from SARIMA-QR model that are plotted against the observed values of sales of banana for the out-sample data. In this figure, the percentage of observed values that lie outside the prediction intervals is around 3%. From this result, it is evident that the prediction intervals of SARIMA-QR model can capture more than the theoretically defined uncertainty (95%). On the other hand, the prediction intervals of SARIMA-MLR model covers nearly 94.5% of the out-sample observations. Unlike the prediction intervals from SARIMA-QR model, it overestimates and underestimates the upper and lower prediction intervals as shown in Fig. 8. In food retail industry, the overestimation and underestimation of prediction intervals will definitely lead to stock-outs and food waste.

The estimators of different quantiles can also be combined to form a robust point estimator. In literature, it was described how to combine the estimators of different quantiles using Tukey's trimean, Gastwirth estimator and five-quantile estimator (Koenker and Bassett, 1978; Taylor, 2007). The robust point estimators are calculated using the above mentioned methods and the results are shown in Table 6.

6.3. Applicability of forecasting models

Each forecasting technique has strengths and weaknesses, and every forecasting situation is limited by resource constraints like data, time, capabilities, and cost. Balancing the merits and demerits of techniques based on constraints and necessities is challenging and important task of management. The applicability of an appropriate forecasting model for a retail industry can be decided based

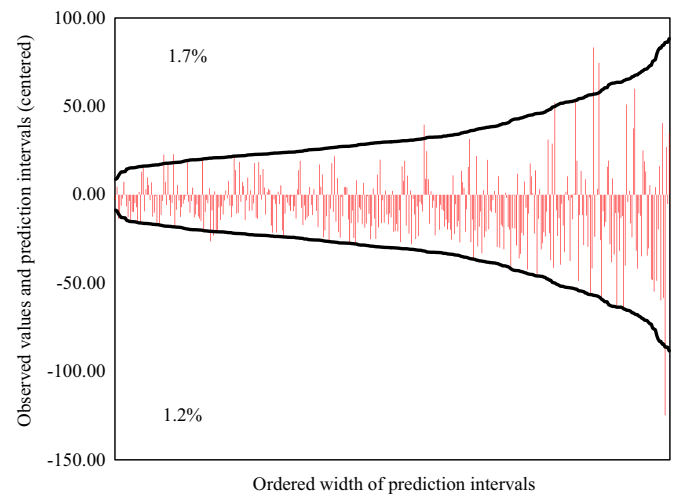


Fig. 9. Graphical representation of 95% prediction intervals from SARIMA-QR for out-sample data.

Table 6
Performance measures for QR estimators.

Robust estimators	MAPE	RMSE
Median ($\tau=0.5$)	23.47	19.35
Trimean	23.54	19.25
Gastwirth	23.44	19.26
Five-quantile	23.62	19.24

on some important technical selection criteria as follows (Armstrong, 2001b; Georgoff and Murdick, 1986; Herbig et al., 1993):

- (1) Data availability: Traditional time series (ARIMA and SARIMA), hybrid seasonal time series (SARIMA-MLR and SARIMA-QR) and MLPNN models proportionately perform better based on the length of historical data (both response and predictor variables). More and detailed the data, superior the accuracy. Comparatively, the MLPNN models require higher amount of data than the time series models for better approximation, i.e., the predictions are less reliable and accurate when the training data are insufficient. On the other hand, the seasonal naïve forecasting model is more suitable when the data of response and predictors variables are qualitative.
- (2) Time series properties: The time series properties (seasonality, trend, and level) are salient features of retail sales. The traditional seasonal time series models (SARIMA) are highly helpful in identifying and explaining any regular or systematic variation due to these time series properties. In addition to the time series properties, the hybrid seasonal time series models (SARIMA-MLR and SARIMA-QR) are also capable to explain the effects of considered demand influencing factors. When there is a strong seasonality and trend, the traditional and hybrid seasonal time series models perform better than the MLPNN models. The MLPNN models can be improved by differencing, and considering autoregressive and moving average elements of time series as input factors. The seasonal naïve forecasting model works fairly well when the seasonality is very much stronger with less variability in the time series. This model does not take into account the trend explicitly, but the variation is so gradual that it will not affect short term retail sales forecast.
- (3) Variability and relationships: Apart from the changes in the company's structure or its atmosphere, the consideration of assumed stable relationships among a model's variable is very important. With the supposition of complete data availability, the MLPNN models are more suitable for explaining the complex non-linear relationships between variables. The SARIMA-MLR model assume linear relationship among variables. But, there is also a possibility that the SARIMA-MLR model would reject an independent variable at mean which may be significant at other quantiles. Unlike the SARIMA-MLR and MLPNN models, the SARIMA-QR model provides more information about relationships between variables at different quantiles. In case of poor knowledge of relationships, the seasonal naïve forecasting model can be chosen.
- (4) Accuracy: In food retail industry, the forecasts are not directly used. Instead, they are transformed into inventory policy, i.e., determining order size based on forecast and safety stock level. Therefore, the forecast inaccuracy can be tolerated or accepted to some level, which must be fixed by the management. As shown in Table 5, the SARIMA-MLR and SARIMA-QR models have better accuracy when compared to MLPNN models. Although the seasonal naïve forecasting model has its simplicity in reflecting the changes in management decisions, its forecast accuracy is not enough to make proper ordering decisions.
- (5) Uncertainty: The first problem of SARIMA-MLR and MLPNN models is that they only provide point forecasts. These models produce only mean forecast, i.e., the forecast will provide only 50% of service level. In addition to this, the SARIMA-MLR model assumes Gaussian distribution for high volatile and skewed time series. As the true distribution of demand or sales is not normal, the extrapolation of higher quantiles from the mean forecast is not going to reflect the reality (Khosravi et al., 2011).

The second problem is the uncertainty in the data due to predictable (e.g. price reductions) and unpredictable (e.g. undesired weather) impacts on the demand. In these cases, the mean forecasts from SARIMA-MLR and MLPNN models may underestimate or overestimate spike or sparse sales, which may lead to stock-outs and food waste respectively. Nonetheless, the forecasts from SARIMA-QR model identifies extreme and sparse sales accurately and directly. It helps the retailer to forecast the higher quantiles (higher service level) directly without using the assumption of normality or quantile extrapolation.

7. Conclusion

In food retail industry, there is a real need of daily sales forecast to support the store managers in precise ordering, without compromising food waste and stock-outs. However, the time series which contain daily sales of perishable food in retail stores, are usually characterized by high volatility and skewness, which are also time varying. These are important constraints, but often ignored in forecasting. In order to overcome these issues, there is a need to develop a time series forecasting model which incorporates uncertainty in forecasts and influence of external variables such day-of-the-week seasonality, month-of-the year seasonality, holidays, festivals, price reduction and weather on the sales. In this study, SARIMA-MLR and SARIMA-QR models are developed and applied to forecast the daily sales of banana in a German retail store. Both of these models yield better predictions for out-sample data, compared to seasonal naïve forecasting, SARIMA, and MLPNN models. Other than this, the derivation of inventory policy from the estimated sales forecast is one of the important problems in stock management in food retail industry. The SARIMA-MLR model produces only the point forecast, i.e., the mean forecast. As the true distribution of demand or sales is not normal, the estimation of prediction intervals from the point forecast is not going to reflect the reality. However, the SARIMA-QR model has additional benefits over the SARIMA-MLR model as follows: (i) it helps to forecast the higher service levels directly and accurately without extrapolation, (ii) the QR results offer detailed and focused insights into the effects of the covariates, and (iii) when the focus of interest is on the higher (for promotional activities) or lower sales (due to extreme weather conditions), the model can aid the management to make accurate and proper decisions.

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Appendix

See Table A1 here.

Table A1
Estimated quantile regression coefficients.

τ	0.05	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	0.95
c	6,2523 [*] (3,1700)	7,2171 [*] (3,1442)	5,7521 ^{**} (1,7359)	4,375 [*] (1,7182)	2,5636 (1,8277)	0,0526 (1,9101)	1,7926 (1,9981)	2,9716 (2,0242)	4,1077 (2,9402)	6,4965 (4,5338)	15,8343 ^{**} (5,6895)
x1	0.4815 ^{**} (0.0471)	0.5388 [*] (0.0287)	0.6306 [*] (0.0304)	0.7154 [*] (0.026)	0.7848 [*] (0.0295)	0.8591 [*] (0.0254)	0.8817 [*] (0.0232)	0.9136 [*] (0.0262)	0.9323 [*] (0.0399)	1.027 ^{**} (0.0204)	1.0164 [*] (0.0587)
x2	−26.1202 ^{**} (8.0065)	−3.7713 (35.6267)	−5.0582 (11.8992)	3.3123 (7.3277)	2.3794 (18.2226)	18.1511 (18.9953)	19.861 [*] (4.493)	17.66 [*] (3.6426)	16.8281 ^{**} (6.1747)	24.5455 [*] (7.5105)	27.4302 ^{**} (2.0007)
x3	20.3775 ^{**} (2.401)	18.9931 ^{**} (4.6767)	19.7617 [*] (10.8491)	24.9049 ^{**} (8.6135)	22.4517 (9.0859)	25.0934 (26.7498)	48.5207 [*] (27.3396)	66.3534 ^{**} (3.0513)	62.5772 ^{**} (5.769)	63.912 ^{**} (5.7458)	54.4522 ^{**} (14.9725)
x4	−2.6648 [°] (1.5945)	−2.2591 (1.3854)	−2.0742 (1.3744)	−2.2551 [*] (0.8919)	−2.006 (1.6469)	0.0836 (1.7276)	−0.4461 (1.5355)	0.5347 (1.5264)	1.5041 (2.3653)	1.7742 (1.8035)	5.8242 (4.4587)
x5	18.5103 [*] (3.3232)	16.8068 ^{**} (1.7072)	16.5579 [*] (7.7551)	20.4982 ^{**} (2.2772)	21.4473 [*] (6.0408)	23.9697 ^{**} (4.5929)	23.4805 ^{**} (7.2013)	32.3404 ^{**} (7.3803)	31.7704 ^{**} (4.6616)	32.8627 ^{**} (11.9606)	32.8868 ^{**} (8.0098)
x6	−3.8311 (3.4139)	−4.9389 (5.5335)	−4.8581 [°] (2.5623)	−6.1033 [°] (3.3412)	−4.8698 ^{**} (1.1495)	−6.3289 ^{**} (1.3216)	−8.3556 [*] (3.2838)	−9.0243 ^{**} (2.5054)	−10.187 [°] (5.745)	−9.9532 (6.7396)	−11.9644 (9.8795)
x7	5.2113 [°] (3.0208)	2.7445 (5.3335)	−0.2873 (2.1432)	−0.7404 (1.1178)	−3.9514 [°] (2.0742)	1.4729 (7.9208)	5.1262 (4.4243)	3.7978 [*] (1.4861)	4.8768 (3.9691)	2.2385 (2.2927)	−0.5889 (8.1299)
x8	−4.8366 (5.1018)	−1.8483 (3.0444)	−0.72 (1.2312)	−3.441 [*] (1.4355)	−2.2982 (2.0394)	0.3202 (2.124)	−0.1432 (1.7994)	−0.4119 (2.1386)	1.1746 (3.3753)	3.3377 (5.235)	−2.8878 (5.2946)
x9	−3.808 (2.5381)	−1.3482 (3.9408)	3.1332 (2.4447)	2.5837 (1.9174)	3.0188 [°] (1.6208)	4.4162 ^{**} (1.5325)	2.9428 [°] (1.6884)	2.7571 (2.2709)	3.2759 (2.7804)	6.9139 (4.8419)	1.8773 (4.7728)
x10	1.054 (2.5321)	−2.3075 (6.6349)	−0.5984 (2.6351)	−0.0168 (2.2969)	1.5384 (3.0244)	3.4662 (2.5813)	4.3587 (3.4018)	7.1659 [*] (3.0313)	7.1004 [*] (3.4237)	9.1252 [*] (5.4933)	6.5214 (5.9796)
x11	8.5293 [*] (2.758)	6.6588 [*] (3.0159)	7.063 [*] (1.8164)	6.5687 [*] (1.9818)	6.7775 [*] (2.1743)	7.7249 [*] (1.934)	7.4363 [*] (2.2535)	8.8462 [*] (2.8248)	9.9449 [*] (2.9989)	11.9246 [*] (5.2676)	6.0798 (5.9463)
x12	5.3691 [*] (2.4431)	2.7808 (3.3759)	4.713 [*] (1.7492)	4.4635 (2.7407)	5.8594 [*] (3.4304)	5.9023 [*] (1.9409)	5.0052 [*] (1.7105)	4.2335 [°] (3.5249)	5.0761 (3.2661)	9.4381 [°] (5.2661)	11.5217 (10.981)
x13	0.4988 (2.6834)	1.5933 (3.0469)	2.7394 [*] (1.3408)	0.3182 (1.4214)	1.225 (2.1205)	2.6088 (2.0246)	2.6412 (1.7818)	2.7979 (2.2192)	3.4878 (2.8025)	3.7409 (4.983)	1.1389 (5.809)
x14	5.0059 (3.423)	3.3039 (3.2334)	2.5942 (1.8033)	2.6746 (2.2397)	2.2819 (2.6225)	2.1795 (2.6017)	2.0319 (2.3881)	0.7667 (3.0767)	4.3268 (4.471)	3.6133 (7.5681)	10.7391 (12.1044)
x15	0.2564 (2.3209)	0.3746 (2.8841)	0.827 (1.3071)	−1.6652 (1.4405)	−1.3395 (1.392)	−1.1793 (1.977)	−0.752 (1.9881)	−1.5381 (1.6907)	−2.8537 (2.9414)	−1.7311 (4.3116)	−10.7143 [*] (5.0964)
x16	−3.6699 (2.4203)	−5.107 [°] (2.9859)	−3.8209 [°] (2.1321)	−2.5824 [°] (1.5357)	−3.8737 ^{**} (1.2116)	−3.7958 [°] (2.0642)	−3.1716 (2.4716)	−3.1617 (2.1981)	1.0884 (5.4868)	4.1708 (4.8644)	−1.4513 (4.8319)
x17	−2.1767 (2.6281)	−4.7951 (2.9222)	−3.2237 (1.9659)	−4.6751 ^{**} (0.9617)	−2.5783 (2.6579)	−0.3486 (2.4301)	−0.1837 (1.689)	−1.2791 (2.2848)	0.5811 (2.5034)	−3.1391 (4.2924)	−9.7355 (6.2711)
x18	−5.6341 (4.128)	−7.6151 ^{**} (2.7916)	−7.9219 ^{**} (2.3312)	−5.7499 ^{**} (2.1666)	−5.4967 ^{**} (1.2831)	−4.9026 ^{**} (1.8001)	−4.6773 [*] (2.1861)	−3.9268 [°] (2.2567)	−2.4746 (3.6837)	0.0232 (4.2161)	−8.0508 (4.8999)
x19	0.7778 (0.7894)	0.7722 [*] (0.1788)	0.5918 ^{**} (0.124)	0.3166 ^{**} (0.1032)	0.4019 (0.1518)	0.5654 (0.3486)	0.529 [*] (0.2253)	0.5924 [*] (0.1433)	1.0469 [*] (0.5794)	1.0383 [°] (0.6087)	0.9403 (0.6544)
x20	0.3429 ^{**} (0.1111)	0.5739 [*] (0.1139)	0.7278 ^{**} (0.085)	0.7365 ^{**} (0.0773)	0.9266 ^{**} (0.1086)	1.0299 [*] (0.0974)	1.1933 ^{**} (0.1153)	1.4307 ^{**} (0.1427)	1.6451 ^{**} (0.0994)	1.6593 ^{**} (0.1648)	2.097 ^{**} (0.2703)
x21	0.0959 (0.1135)	−0.0231 (0.1056)	−0.1569 (0.1207)	−0.0283 (0.1525)	0.1017 (0.151)	0.0777 (0.1328)	0.0762 (0.0464)	−0.0526 (0.124)	0.0643 (0.2435)	−0.4153 [*] (0.2109)	−0.2675 (0.2411)
x22	−0.1963 [°] (0.1022)	−0.2373 [°] (0.1402)	−0.2944 ^{**} (0.0909)	−0.1954 [°] (0.1059)	−0.1353 (0.1052)	−0.1623 (0.1124)	−0.2166 [*] (0.107)	−0.2389 [*] (0.1211)	−0.0153 (0.1859)	−0.2951 [*] (0.1234)	−0.2772 (0.1849)
x23	−1.233 [*] (0.4035)	−0.7351 (0.9812)	−0.3282 [*] (0.1508)	−0.2418 (0.1764)	−0.1743 (0.2917)	−0.1445 (0.1671)	−0.3549 [*] (0.1599)	−0.2537 (0.1795)	−0.5081 (0.508)	0.1451 (0.2001)	−0.1285 (0.1476)
x24	0.0449 (0.0867)	0.0242 (0.0522)	0.0748 (0.0521)	0.1149 ^{**} (0.0238)	0.0983 ^{**} (0.0337)	0.1075 ^{**} (0.0382)	0.1101 ^{**} (0.0257)	0.1128 ^{**} (0.043)	0.1522 [*] (0.0704)	0.0798 (0.0563)	0.1286 ^{**} (0.0424)

Standard error estimates in parenthesis.

[°] Significance level: ≥ 0.1 .

^{*} Significance level: ≥ 0.05 .

^{**} Significance level: ≥ 0.01 .

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